

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

**ATLAS: ADVANCED THESIS LOGISTICS AND AI SYSTEM FOR THE
COLLEGE OF ENGINEERING, COMPUTER STUDIES AND
ARCHITECTURE AT LYCEUM OF THE PHILIPPINES UNIVERSITY -
CAVITE**

An Undergraduate Thesis
Submitted to the Faculty of the
College of Engineering, Computer Studies and Architecture
Lyceum of the Philippines University-Cavite

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in Computer Science
with specialization in Software Engineering

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COLLEGE OF ENGINEERING, COMPUTER STUDIES, AND ARCHITECTURE

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ACCESS LEAF

**Bachelor of Science in Computer Science
with Specialization in Software Engineering**

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**ATLAS: ADVANCED THESIS LOGISITICS AND AI SYSTEM FOR
COECSA OF LYCEUM OF THE PHILIPPINES UNIVERSITY CAVITE**

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ACCEPTANCE SHEET

This thesis entitled **ATLAS: ADVANCED THESIS LOGISTICS AND AI SYSTEM FOR LPU-CAVITE COECSA DEPARTMENT** prepared and submitted by **WINSTON AGUSTIN, NEIL VICEDO, JERALD RYAN D. GERONA, and IVAN KERWIN ILANO**, in partial fulfillment of the requirements for the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH SPECIALIZATION IN SOFTWARE ENGINEERING** is hereby accepted.

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LIST OF ABBREVIATIONS

ATLAS	Advanced Thesis Logistics and AI System
LPU-C	Lyceum of the Philippines University Cavite
COECSA	College of Engineering, Computer Studies, and Architecture
LLM	Large Language Model
SQL	Structured Query Language
CMS	Content Management System
GA	Genetic Algorithm
ASS	Automated Scheduling System
DSS	Decision Support System
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
PHP	Hypertext Preprocessor
NLP	Natural Language Processing

ABSTRACT

Thesis defense management and evaluation processes within the academic institution are often challenged by scheduling conflicts, inconsistent assessments and limited tools to ensure originality in research titles. This study aims to develop ATLAS (Advanced Thesis Logistics and AI System), a web-based system that utilizes AI to streamline scheduling, improve assessment accuracy, and promote academic integrity within the College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University Cavite (LPU-C). ATLAS integrates the Genetic Algorithm for an optimized scheduling system and Gemini, a Large Language Model (LLM), to aid in plagiarism detection and sentiment analysis for thesis feedback. Data from student researchers, thesis advisers, and program chairs within COECSA were collected to evaluate system effectiveness based on ISO/IEC 25010 quality standards, covering functionality, usability, and reliability.

Keywords: thesis defense management, web-based system

CHAPTER I

INTRODUCTION

Background and Rationale of the Study

The management of thesis processes within academic institutions is increasingly recognized as a vital component of education success. As the demands on students grows, particularly in higher education, the need for effective thesis management systems has become paramount. These systems not only streamline the administrative aspects of thesis supervision but also enhance the overall academic experience for both students and faculty. The significance of thesis management systems lies in their ability to facilitate timely completion of thesis, improve communication between students and advisers, and provide structured support throughout the thesis writing process (Abidin, 2023) nasakin link neto lagay ko nalang kung final na to

Thesis management systems play a crucial role in fostering student engagement and academic performance. By offering tools for tracking progress, managing deadlines, and providing resources for research, these systems empower students to take ownership of their academic work. Furthermore, the integration of technology into thesis management allows for real-time feedback and monitoring, which is essential for maintaining the quality of academic outputs. PA CHECK :D

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	Thesis management involves the coordination of research presentations, grading, and documentation within academic institutions. As student populations grow, manual processes for scheduling defenses, evaluating theses, and organizing academic records become increasingly inefficient. These traditional methods lead to time conflicts, delays in feedback, and inconsistencies in evaluation. Incorporating AI-driven systems can address these challenges, ensuring streamlined processes and improving overall transparency. The College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University-Cavite (LPU-C) is committed to empowering students with practical skills, fostering lifelong learning, and promoting global leadership and social responsibility. With outcomes-based education, COECSA achieves above-national-passing rates in board exams for its programs, including Computer Science, Information Technology, Engineering, and Architecture. The college holds notable accreditations, such as Level 3 PACUCOA status for BS Information Technology and Level 2 for engineering programs and provides certifications like MTA and CCNA to ensure industry readiness. Guided by a vision to lead in engineering, architecture, and ICT education, COECSA strives to produce quality-centered graduates who uphold truth, fortitude, and service to God and country, while actively contributing to community development through innovative programs. The College of Engineering, Computer Studies, and Architecture (COECSA) of Lyceum of the Philippines University-Cavite (LPU-C), faces logistical challenges in managing research presentations for its growing student	
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	population. The current manual system for defense schedules, evaluating theses, and organizing documents results in inefficiencies, time conflicts, and delayed feedback. The increasing demand for streamlined academic processes highlights the need for an AI-driven system to optimize scheduling, automate assessments, and enhance overall transparency. Currently, Lyceum of the Philippines University-Cavite (LPU-C) is using manual system for the managing capstone and research presentations. This traditional process involves time-consuming tasks such as scheduling, which requires coordinating the availability of students, advisers, and panelists, and often results in conflicts or delays. The grading system, similarly, relies on manual input from the panelists, making it prone to errors and inefficiencies. The management of research projects, from recording schedules to tracking grades, also lacks a centralized, automated platform, further complicating the process as the student population grows. Managing academic content effectively is a critical component of streamlining thesis logistics. To address these challenges, the study proposes the development of ATLAS: Advanced Thesis Logistics and AI System was developed for the College of Engineering, Computer Studies, and Architecture (COECSA) at Lyceum of the Philippines University-Cavite (LPU-C). ATLAS integrates a suite of AI-driven tools designed to streamline the thesis journey from topic selection to final evaluation. The Thesis Topic Decision Tool assists students in identifying relevant and impactful research topics by leveraging AI algorithms that analyse current academic trends, aligning student interests with high-impact areas of study. To ensure originality, the Research Title Acceptance Tool uses the Gemini Pro Large	
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	Language Model (LLM) to assess the uniqueness of proposed thesis titles, preventing duplication and fostering innovative research. The Scheduling System, powered by a Genetic Algorithm, automates the complex process of coordinating defense schedules, reducing conflicts and optimizing the availability of students, advisers, and panelists. Additionally, the Content Management System (CMS) offers a centralized platform for managing academic resources, such as rubrics and panel assignments, streamlining administrative tasks and enhancing transparency. The Requirement Checker Tool aids in tracking the submission of mandatory documents. Finally, the Pass Recommendation System employs AI-based sentiment analysis to evaluate feedback from panelists, providing consistent and objective recommendations while maintaining human oversight. Together, these integrated tools address the critical pain points in the thesis management process, promoting efficiency, accuracy, and academic integrity. Research supports the necessity for an AI-driven system, revealing numerous important challenges in academic processes. Dajda et al. (2021) discovered that students frequently find it challenging to choose relevant thesis topics that align with current research trends, which often results in producing less impactful research. It highlighted the increasing importance of guiding students towards popular and high-impact areas like neural networks, machine learning, and software engineering practices. The research noted that approximately 25% of the analyzed bachelor projects incorporated machine learning algorithms, reflecting a shift towards data-driven solutions. Furthermore, the study emphasized the value of aligning academic projects with industry-relevant technologies to enhance the	
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	practical skills of graduates. This trend underscores the critical need for educational institutions to update their curricula to support these evolving industry demands. Nasien et al. (2022) discusses how Genetic algorithms are highly effective in course scheduling, especially in environments with complex constraints like time slots, room availability, and instructor schedules. The study highlights that genetic algorithms optimize schedules by iteratively evolving solutions through selection, crossover, and mutation processes, thus consistently improving schedule quality and minimizing conflicts. This approach enables the creation of a well-balanced schedule by ensuring that instructors are not double-booked and that room assignments avoid overlap. Additionally, the flexibility of genetic algorithms allows for adaptation to different scheduling requirements across semesters, making them suitable for dynamically changing academic settings. This study affirms that using genetic algorithms in academic scheduling significantly enhances efficiency, reduces manual scheduling efforts, and leads to better resource utilization. This methodology not only reduces the time and effort required to generate schedules but also enhances the quality of outcomes. The effectiveness of GAs in this context has been demonstrated by achieving higher satisfaction ratings from students and staff, as the scheduling system adapts dynamically to complex constraints and fluctuating inputs. LAST NA GAGAWIN TO	
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	Objectives of the Study The study aims to develop ATLAS: Advanced Thesis Logistics and AI System for the College of Engineering, Computer studies, and Architecture at Lyceum of the Philippines University - Cavite. Specifically, the project will: 1. Develop ATLAS: Advance Thesis Logistics and AI System for LPU-Cavite COECSA Department with the use of the following features. a) Thesis Topic Decision Tool • This tool utilizes an API or an internal database to suggest relevant research topics based on current academic trends, ensuring quality thesis projects. b) Research Title Acceptance Tool • This tool assesses the uniqueness of a title using sources from open-access repositories and analyses them with the Gemini Pro LLM Model. c) Scheduling System • Aligns the defense schedule of both students and panelists based on availability. A feature which uses the Genetic Algorithm. d) Content Management System • Allows modular modification of rubrics, team lists, title lists, and the panelists assignments and the overall looks of the system.	
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	e) Requirement Checker Tool • Helps students and advisers track the completion of necessary documents before the defense. f) Pass Recommendation System • Utilizes the Gemini Pro LLM to process scores, comments, and the paper's sentiment of the panelists to generate automated, suggestive feedback, improving consistency and reducing errors. 2. The project will use HTML/CSS and JavaScript for front-end flexibility and a user-friendly interface. PHP will be used for backend to manage server-side logic and real-time data processing. For database management, MySQL will be utilized to provide scalable and efficient data storage. 3. Evaluate and improve the system using ISO/IEC 25010 and Mobile Application Rating Scale (MARS). ISO/IEC 25010 a) Functional Suitability b) Performance Efficiency c) Compatibility d) Interaction Capability e) Reliability f) Security g) Maintainability	
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h) Flexibility

i) Safety

Mobile Application Rating Scale

a) Engagement

b) Functionality

c) Aesthetics

d) Information Quality

Significance of the Study

The development of ATLAS: Advanced Thesis Logistics and AI System for LPU-Cavite COECSA Department aims to change how research presentations are scheduled, evaluated and managed. This study is important as it introduces an AI-driven system that rationalizes the logistical challenges faced by both students and school faculty in organizing thesis defenses, while improving the overall efficiency and transparency of the academic process.

Student Researchers of LPU-C will greatly benefit from this system. ATLAS offers students clear and organized scheduling, performance assessments, and presentation requirements. Our system improves the research and defense processes accountability and transparency.

Thesis and Capstone Advisers will also experience significant advantages. ATLAS will provide them an efficient platform for scheduling optimizing time management, research presentations, and minimizing conflicts. Using our system,

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	advisers may monitor the students' area of improvement and make well informed decisions. Research Teachers of LPU-C COECSA will benefit from the system's AI capabilities in facilitating research title validation, plagiarism detection, and communication between students and faculty. By implementing our system LPU-C's commitment on using modern technologies to improve academic processes. Future Researchers stand to benefit from this study as it can serve as a model for incorporating AI into academic scheduling and assessment systems. Future Researchers could utilize this framework to improve and innovate the academic scheduling and evaluation system. Scope and Limitation This study is centered on the development and implementation of ATLAS: Advanced Thesis Logistics and AI System, specifically customized for the College of Engineering, Computer Studies and Architecture (COESCA) at Lyceum of the Philippines University-Cavite (LPU-C). The system includes the following key features such as Scheduling System which automates the scheduling of thesis defenses to minimize conflicts and improve efficiency, Pass Recommendation Tool that utilizes sentiment analysis via Gemini, to analyze feedback and suggest pass/fail outcomes, Research Title Acceptance that detects similarities between proposed and existing research titles to prevent plagiarism and ensure originality, Requirement Checker Tool that tracks thesis chapter submissions and ensures compliance with deadlines and academic requirements, and lastly Thesis Topic	
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	Decision Tool that provides students with topic suggestions based on trends in computing and AI. The research will take place at LPU-C and will concentrate on overcoming the logistical and academic challenges faced by COESCA, with data collection being carried out over one semester. The system will be applicable only within COESCA, with its features specifically designed to meet the needs of the department. These features may require modifications for use in other departments. The study will include key stakeholders such as students, research advisers, and panelists. It will not address broader administrative activities, plagiarism checks beyond title comparisons, or other academic programs at the university. Although AI-driven tools like the Gemini Pro LLM will aid in the decision-making process, human input is important for final grading and evaluation to maintain fairness and accuracy. The system necessitates an internet connection and compatible devices, and it is built for web-based access. ATLAS requires a reliable internet connection, and it requires a Windows 10 operating system version 10.0.19045 Build 19045. For CPU a minimum of Intel I3 3rd Gen or Ryzen 3 series or 4000 series APU and a minimum of 4gb RAM. As for the resolution, a minimum of resolution of 1280 x 720 (also with a 16:9 aspect ratio). For browsers, Google Chrome 131.0.6778.109 (Official Build) (64-bit), Opera version LVL6 (core: 114.0.5282.233), Microsoft Edge version 131.0.2903.86 (Official build) (64-bit), and Firefox version 131.0.2 (64-bit).	
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For mobile users, the minimum Android Version is Android 10, for iOS is iOS 17. As for the resolution for mobile devices, the system requires a minimum resolution of 720 x 1280 (with a 16:9 aspect ratio) to ensure optimal functionality and a fully responsive design. These specifications allow users to effectively use all features, with a layout designed for both mobile and desktop environments.

CHAPTER II

REVIEW OF RELATED LITERATURE

The proponents of this project collected relevant literature and studies to facilitate the research. This collection aims to assist readers in understanding the study's content. The literature review clarifies terms closely linked to the research, while the review of related studies delves into findings and recommendations important to this project. Also, a synthesis is provided at the end of each topic to show the connection between the literature and the researcher's work.

Genetic Algorithm

Genetic Algorithms are powerful optimization techniques inspired by the principles of natural selection and genetics. Since their inception by Holland in the 1970s, GAs have evolved into versatile tools widely applied across various domains, including engineering, artificial intelligence, and computational biology. The fundamental mechanism of Genetic Algorithm involves iteratively searching for optimal solutions through processes analogous to biological evolution, such as selection, crossover, and mutation (Chen and Okle, 2024). Genetic Algorithms have garnered significant attention in recent years due to their versatility and effectiveness in solving complex optimization problems across various fields. Their

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	application ranges from machine learning to biomedical research, showcasing their adaptability and robustness in diverse scenarios. One prominent application of Genetic Algorithm is in the optimization of machine learning models. For instance, Wang (2023) developed an automatic grading system for English compositions that integrates a Genetic Algorithm with a Convolutional Neural Network (CNN). This approach enhances grading accuracy and optimizes the evaluation process by continuously refining the model through genetic operations such as selection, crossover, and mutation (Wang ,2023). Genetics Algorithm are used within ATLAS for optimization tasks, particularly in enhancing scheduling and resource management. Inspired by natural selection, genetic algorithms iteratively search for the best solutions to complex problems. In ATLAS, they can be applied to optimize schedules, ensuring that defense presentations are arranged with minimal conflicts by efficiently allocating time slots based on the availability of students, advisers, and panelists. This not only reduces scheduling conflicts but also improves the overall efficiency of the system, making it more adaptable to the growing number of students and faculty involved in research. Large Language Model (LLM) Large Language Models (LLMs) have emerged as transformative tools in various fields, particularly in Natural Language Processing (NLP). Their architecture, characterized by a vast number of parameters, allows them to capture the complexities of human language, enabling applications ranging from text	
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generation to complex problem-solving in scientific domains. For instance, recent studies have highlighted the application of LLMs in gene-related tasks, for instance, recent studies have highlighted the application of large language models in gene-related tasks, where these models are used to enhance gene-focused analysis through retrieval-augmented generation techniques, showcasing their versatility beyond traditional text processing (Lin, 2024).

The computational demands of LLMs, however, pose significant challenges. As noted in recent research, while these models achieve unprecedented performance levels, they require substantial computational resources for both training and inference, which can limit their accessibility and practical applications (Huangpu, 2024). This has led to ongoing efforts in model compression and knowledge distillation, aimed at maintaining performance while reducing the computational burden (Huangpu, 2024). Moreover, the scalability of large language models, as demonstrated by the increasing size of models like PaLM, highlights the need to optimize their architectures to improve their practical effectiveness across various applications (Cai, 2023).

The study highlights the ethical and academic challenges of students using Large Language Models (LLMs) to write theses, including concerns over academic integrity, lack of authentic learning, and the risk of passing off AI-generated work as their own. While guidelines were proposed to address these issues, our thesis improves on this by integrating AI responsibly. Through tools like the Thesis Topic Decision Tool and Research Title Acceptance Tool, our system supports students in selecting relevant and unique research topics while ensuring originality.

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	Additionally, the Pass Recommendation System ensures that AI aids decision-making without replacing critical human evaluation, safeguarding both academic integrity and learning outcomes. (Leary ,2023) LLM such as the Gemini Pro LLM used in ATLAS, adds a layer of AI-driven intelligence for thesis evaluations and feedback. LLMs can analyze text data, such as research titles or student-submitted documents, to provide relevant feedback, detect potential issues, and suggest improvements. For instance, the LLM helps in assessing the uniqueness of thesis titles by comparing them with existing works and generating recommendations to avoid redundancy or plagiarism. The use of LLM in ATLAS ensures that evaluations are more consistent and precise, as it supports the decision-making process with AI-driven insights, while human reviewers still maintain control over the final evaluations. Thesis Management System The development of thesis management systems has become increasingly relevant in the context of higher education. The integration of information technology into academic processes has been shown to enhance the quality and efficiency of thesis supervision and completion. Tiwow (2023) emphasizes that aligning thesis projects with the actual needs of the IT industry can make student research more relevant and impactful. By implementing effective time management programs and project management training, students can approach their thesis projects with greater confidence and depth. This technological	
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	integration not only aids in project execution but also fosters a more engaging learning environment. In addition, the introduction of specialized thesis management systems, such as Penamasy, has been proposed to streamline the thesis writing process. Semma (2023) presents this system as a solution to the significant challenges students face during thesis writing, which often leads to stress and confusion. By providing structured support and resources, such systems can alleviate some of the burdens associated with thesis completion. The literature highlights the critical role of thesis management systems in enhancing academic performance. The integration of technology is essential for fostering a positive environment for thesis completion. As institutions continue to navigate the complexities of modern education, the development and implementation of effective thesis management systems will be pivotal in supporting students' academic journeys. Automated Scheduling System The development of automated scheduling systems has gained significant attention in recent years, particularly with advancements in machine learning and artificial intelligence. The integration of Automated Scheduling Systems (ASS) in educational institutions has gained significant traction in recent years, driven by the need for efficiency and effective resource management. These systems leverage advanced algorithms and artificial intelligence (AI) to streamline the scheduling process, thereby alleviating the administrative burden on educators and enhancing	
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	the overall educational experience for students. One of the primary advantages of Automated Scheduling Systems is their ability to optimize class schedules based on various constraints such as classroom availability, instructor schedules, and student preferences. For instance, the development of an automated scheduling system using the 1C:Enterprise platform has demonstrated significant improvements in managing educational processes within higher education institutions. This system not only reduces the time required for scheduling but also facilitates easier modifications to the timetable, ensuring that the educational process runs smoothly (Rymar & Voytko, 2021). Similarly, research has shown that employing genetic algorithms can effectively address complex scheduling problems in universities, allowing for the generation of optimal course schedules that meet diverse constraints (Lei et al, 2024). In addressing the challenges of scheduling in universities, (Chen et al., 2021) focused on optimizing class schedules through automated conflict resolution and efficient resource management. Their work highlights the importance of minimizing scheduling conflicts to enhance the educational experience for both students and faculty. While their study employed traditional software algorithms to achieve these goals, our system applies AI-driven scheduling algorithms to further enhance the accuracy and efficiency of the scheduling process. This approach not only streamlines the scheduling of research presentations but also optimizes panelist and student availability, offering a more flexible and responsive solution to scheduling challenges.	
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	Automated Scheduling System is another critical component that simplifies the logistics of research presentations and defenses. By leveraging AI and algorithms, this system automatically aligns the schedules of students, advisers, and panelists, ensuring that conflicts are minimized and that presentations can proceed smoothly. It considers various constraints, such as availability, deadlines, and institutional requirements, to generate the most efficient and conflict-free schedule possible. This automated process reduces the manual workload on administrators and minimizes errors, making it easier to manage multiple research defenses in a growing academic environment. Decision Support System Decision Support System is a computer program that helps decisions, assessments, and actions inside a company or organization. Massive data sets are sorted through and analyzed by a Decision Support System, which then compiles detailed information that may be utilized in decision-making and problem-solving. The theory and practice of decision support systems are progressing and occasionally intersecting with other notable breakthroughs in information technology, including organizational computing, e-commerce, and pervasive computing (Abdul-kareem et al., 2023). Decision Support System is essential in guiding students and faculty members throughout the thesis process by providing structured recommendations based on analyzed data. In the context of ATLAS, the DSS helps students select thesis topics that align with current academic trends and their strengths. By	
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	and a more robust content management system. These enhancements will further streamline the process, reduce conflicts, and provide intelligent assistance for both faculty and students CMS will serve as a crucial tool for managing academic content within ATLAS. It provides a flexible platform for faculty members to easily update rubrics, research titles, team lists, and panelist assignments without needing technical expertise. The CMS ensures that all important documents and data related to thesis projects are well-organized and easily accessible. By centralizing content management, the CMS will help streamline academic processes, reduce manual errors, and enhance the overall efficiency of handling research presentations and evaluations. Generative Artificial Intelligence Artificial Intelligence (AI) has emerged as a transformative force across various sectors, influencing methodologies, operational efficiencies, and ethical considerations. In the realm of education, AI's potential to enhance teaching and learning processes is increasingly recognized. Research indicates that AI can significantly improve educational management by optimizing administrative tasks and personalizing learning experiences for students and teachers alike (Pisica et al., 2023). Studies have shown that the integration of AI in higher education can lead to improved teaching methodologies and learning outcomes, although challenges remain in its implementation (Begum, 2024).	
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	AI is not merely a technological advancement but a catalyst for profound changes across various sectors. Its applications in education emphasize the need for ongoing research into the ethical, social, and economic implications of AI technologies. As AI continues to evolve, it is essential to foster interdisciplinary collaboration to address the challenges and harness the opportunities presented by this transformative technology. Artificial Intelligence in Education AI-powered adaptive learning platforms have shown promising results in improving student performance, particularly in subjects like mathematics. For instance, a study demonstrated that these platforms positively influenced students' mathematical skills, indicating that AI can effectively tailor educational experiences to individual learning needs Dabingaya (2022). This aligns with findings from other research that emphasizes the role of AI in personalizing learning pathways and providing real-time feedback, which can significantly enhance student engagement and achievement (Trang, 2024). The application of Artificial Intelligence (AI) in research has gained significant traction in recent years, particularly in the context of academic writing, data analysis, and educational methodologies. This literature review synthesizes recent studies that highlight the transformative potential of AI across various research domains, emphasizing its role in enhancing academic performance, improving research methodologies, and addressing ethical considerations. AI's integration into academic writing has been a focal point of recent research. For	
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	instance, William (2024) discusses how AI is reshaping academic research, particularly in personalized learning and intelligent tutoring systems, which bridge the gap between academic results and industry needs. This sentiment matches with Pigola (2023), who notes that AI's influence extends beyond mere data processing to fundamentally altering the landscape of academic research itself. The potential of AI tools, such as Gemini, to assist in literature reviews and data analysis has been recognized as an asset for researchers, facilitating more efficient and thorough investigations (Burger et al., 2023). Furthermore, the exploration of AI's role in enhancing students' educational performance underscores its capability to tailor educational content based on individual learning patterns, thereby improving engagement and knowledge retention (Abbas et al., 2023). The integration of AI in education presents a multifaceted opportunity to enhance learning experiences and outcomes. As educational institutions continue to adopt AI technologies, ongoing research and collaboration will be essential to navigate the complexities and maximize the benefits of AI in educational contexts. Evaluating Academic Systems Using ISO/IEC 25010 Standards The evaluation of academic systems using the ISO/IEC 25010 standards has gained prominence in recent years, as educational institutions increasingly rely on software solutions to enhance their operational efficiency and educational outcomes. ISO/IEC 25010, part of the Software Quality Requirements and Evaluation (SQuaRE) framework, provides a comprehensive model for assessing	
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	software quality through eight key characteristics: functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, flexibility, and safety (Ani et al., 2022). One significant study by Mulyawan et al. (2021) emphasizes the importance of ISO/IEC 25010 in assessing the quality of academic information systems. The authors argue that the standard's comprehensive nature allows for a nuanced evaluation of software performance in educational settings, ensuring that systems meet the diverse needs of users, including students and faculty. This aligns with findings from Rahmad (2023), who demonstrated that integrating ISO/IEC 25010 into the "PSG Balita" application improved the quality of nutritional status data by streamlining data management processes, thereby enhancing the overall educational experience. Such improvements underscore the potential of ISO/IEC 25010 to facilitate better data handling and reporting in academic contexts. The ISO/IEC 25010 standards serve as a robust framework for evaluating academic systems, offering a structured approach to assess software quality in educational environments. The integration of these standards not only enhances the functionality and reliability of educational software but also contributes to improved user satisfaction and learning outcomes. As educational institutions continue to adopt advanced technologies, the relevance of ISO/IEC 25010 in ensuring quality and effectiveness in academic systems is likely to grow.	
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Conceptual Framework

The proponents undertaking the project must acquire knowledge in HTML, CSS, JS, SQL, and PHP, as these skills are essential for the development of Decision Support System. The framework shows the system's components and their relationship, focusing on how an AI-driven website processes can optimize scheduling and assessment within the academic environment.

Figure 1. *Conceptual Framework of ATLAS.*

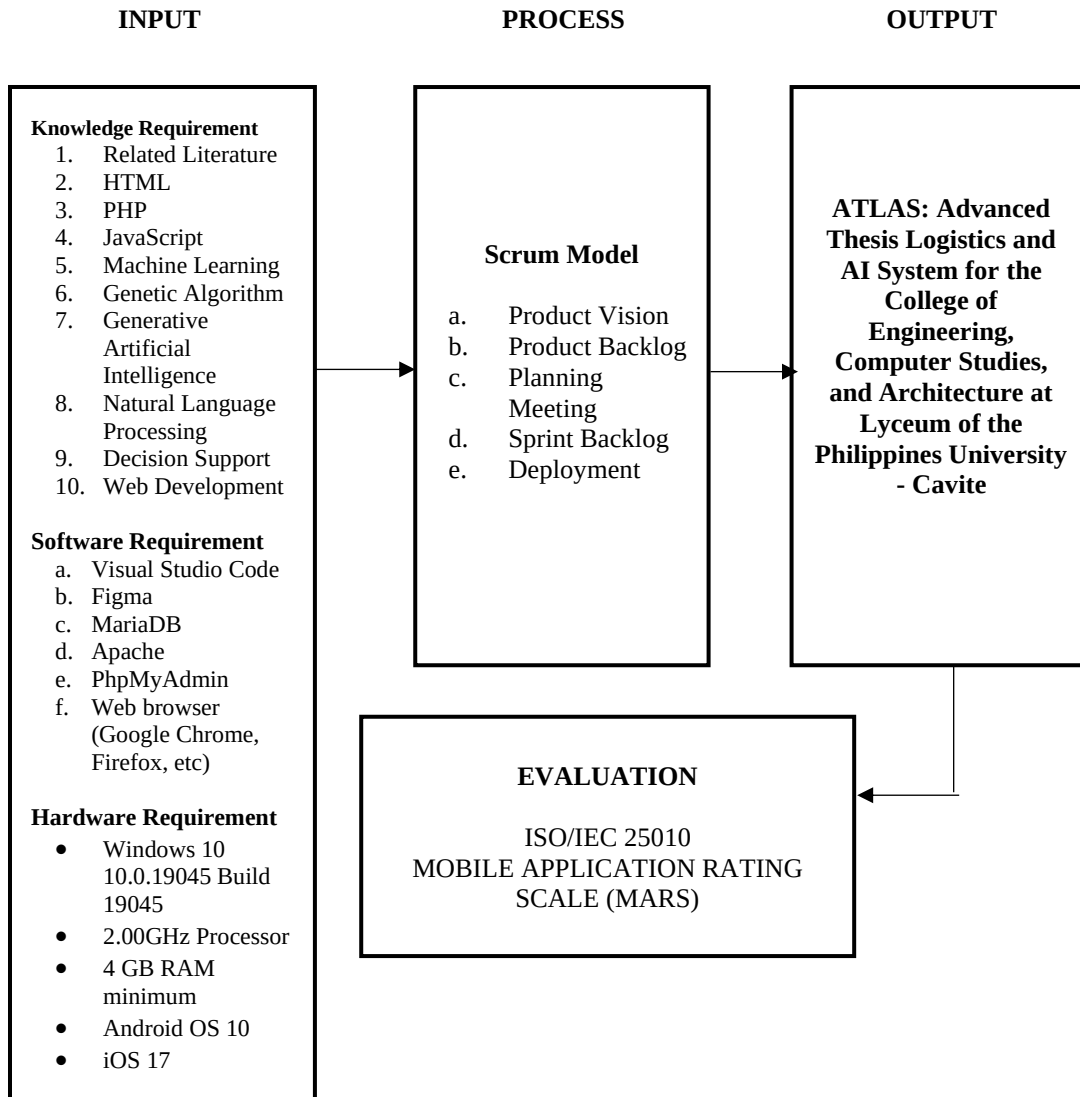


Figure 1 illustrates the project's framework representing the Input, Process, and Output phases of the study. The Input section shows the required Knowledge requirements, Software requirement, and Hardware requirements which are necessary for the proponents to develop the system. Next is the Process section, which the proponents follow throughout the development process of the system.

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	Finally, the Output section, which shows the outcome of the study, an AI-Driven System. Once the system has finished its development, it will undergo an evaluation using the ISO/IEC 25010 and MARS standard to evaluate its effectiveness and quality. Definition of Terms For the readers to fully comprehend this study, the following are the notable terms that have been utilized in the study: ATLAS. Advanced Thesis Logistics and AI System is an AI-powered system designed to automate academic processes like scheduling, data analysis, and decision-making for research presentations. Figma. A web-based design tool that allows users to create, share, and test designs for websites. Figma is used to create the wireframe of ATLAS. MySQL. A relational database management system that stores and manages data. In ATLAS, MySQL is used to store tables and data from the system. Content Management System. A digital platform that helps users create, manage, and publish content for websites and apps. In ATLAS it streamlines document management, ensures transparency, reduces redundancy, and minimizes	
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	errors, providing students and faculty with easy access to essential materials for a cohesive academic experience. Gemini. A Large Language Model (LLM) developed by Google AI. In ATLAS, Gemini is used as it empowers the Thesis Topic Decision Tool, Research Title Acceptance, and Pass Recommendation features. Large Language Model. A type of AI program that uses deep learning to process and generate human language. LLM allows the features in ATLAS to work. Genetic Algorithm. A computational method that uses natural selection and genetic operators to solve problems by evolving a population of candidate solutions. In ATLAS, GA optimizes scheduling by efficiently assigning research presentations to available slots.	
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CHAPTER III

METHODOLOGY

Research Design

In this study, the proponents used a quantitative survey research approach to evaluate and see the potential effectiveness of the ATLAS: Advanced Thesis Logistics and AI-System for COECSA of Lyceum of the Philippines University Cavite. This approach was chosen to get valuable data from students, advisers, and program chairs. The data will be analyzed to see and understand the relevance of the proposed system.

To optimize scheduling, the Genetic Algorithm was chosen as the main algorithm for ATLAS. Genetic Algorithm is a robust optimization technique inspired by the process of natural selection. It is used in ATLAS to efficiently allocate limited slots for research presentations and to avoid conflicts, ensuring that all users of ATLAS have an optimized and organized schedule.

Additionally, Gemini, a Large Language Model (LLM), is utilized within ATLAS to perform tasks related to natural language processing (NLP). Gemini aids in the text analysis of ATLAS, especially in sentiment analysis of feedback from the panelists, plagiarism detection in research titles, and suggesting thesis topics with emerging trends in computing.

Sampling Technique

Genetic algorithms (GA) are especially effective for scheduling because of their flexibility and ability to navigate extensive solution spaces with detailed constraints, surpassing conventional heuristic algorithms such as simulated annealing, expert systems, and ant colony algorithms. While these alternative methods can be useful, they frequently encounter challenges in addressing the multi-dimensional and energetic constraints fundamental in scheduling problems. For example, simulated annealing depends on a cooling schedule, which can be difficult to optimize for complex scheduling scenarios. In contrast, expert systems necessitate comprehensive rules that may not easily adapt to energetic alterations in scheduling constraints. Although the ant colony algorithm is successful in identifying paths in discrete optimization problems, it generally requires substantial computational resources to manage large scheduling spaces and may exhibit slower convergence. By contrast, the GA approach iteratively evolves solutions using adaptive operators (selection, crossover, and mutation) to enhance solution diversity and avoid local optima, making it highly effective for global optimization in scheduling. This study demonstrates that the improved genetic algorithm, enhanced with a local search operator and adaptive crossover and mutation rates, achieves faster convergence and higher-quality solutions. The experiments show that, compared to other algorithms, the GA yields better average fitness values and computational efficiency, making it an ideal choice for addressing the multi-objective demands of course scheduling. (Xu., 2021)

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	Gemini, with its dual encoder model and memory-augmented mechanism, is uniquely positioned to excel in thesis management compared to other AI models. Unlike traditional algorithms, which may lack the ability to handle complex, multi-turn dialogues, Gemini’s architecture is specifically designed to understand and engage in sequential, context-rich interactions. This enables it to navigate the nuanced and evolving steps in thesis tasks, such as suggesting relevant topics, validating research titles, and providing pass recommendations. The model’s memory-augmented mechanism allows it to retain context across multiple dialogue turns, making it ideal for tasks that require consistency and continuity, such as following up on a student’s thesis development over time. Additionally, Gemini’s fine-grained dialogue state tracking ensures accurate and relevant responses by keeping track of the evolving context, which is essential in academic environments where clarity and adherence to specific guidelines are critical. While other Large Language Models (LLM) offers strengths in open-ended conversation and task-specific support, Gemini’s tailored approach to multi-step dialogue and task tracking makes it a superior choice for the specialized requirements of thesis management. (Krify, 2024) We chose Gemini over GPT because it meets the specific technical requirements of our project, particularly in handling multi-turn dialogues and providing context-aware interactions tailored for academic processes. Gemini's memory-augmented mechanism ensures it can retain and utilize context across multiple interactions, making it highly effective for complex tasks like thesis topic suggestions and sentiment analysis. Additionally, its architecture is optimized for	
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	fine-grained dialogue state tracking, enabling more accurate and relevant responses, which are critical for guiding students through nuanced academic workflows. Also, in an article by Jackson (2023) it is stated that Google AI is one of the ethical companies that practices AI. Stratified random sampling will be employed to gather responses from Professors, Research Advisers, Research Panelists, Department Chairs, and Students through an online survey. This approach involves dividing the population into distinct subgroups (strata) based on their roles within the thesis process. A random sample will then be drawn from each subgroup, ensuring a proportionate and representative selection of participants. By capturing input from all relevant groups, this method ensures comprehensive feedback on the college’s thesis system and the potential benefits of an AI-driven decision support tool, while minimizing bias and enhancing the validity of the research findings. Participants of the Study The participant of this study includes three key groups from LPU-C COECSA Department: students, thesis advisers/professors, and the program chair. All the mentioned participants are vital in giving valuable feedback on potential improvements of ATLAS. • Student Researcher/Proponent will provide insights on how the system supports their scheduling, performance tracking and research title management	
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- Research advisor monitoring the students' progress, and improve the evaluation of research presentations
- Research Panel will assess the student's performance during defense.
- The Program Chair will offer its insights into the system's overall impact on the department.

Research Locale

This study will be conducted at Lyceum of the Philippines University – Cavite (LPU-C), specifically within the College of Engineering, Computer Studies, and Architecture (COECSA), located in General Trias, Cavite. LPU-C is known for its commitment to academic excellence and fostering innovation, making it an ideal setting for this study.

COECSA Department has been chosen as it houses more complex research presentations and academic assessments. The department is home to a diverse group of faculty members and students who actively engage in academic research, making them the main beneficiary of the proposed system. By focusing on COECSA, the study aims to address real-world challenges in managing research defenses and enhance the efficiency of academic assessments.

Research Instrument

To gather data for this research, an online survey questionnaire using Microsoft Forms. The survey will target the key stakeholders including Professors,

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	Research Advisers, Research Panelists, Department Chairs, and Students. The objective is to gain valuable insights into the current thesis system used within the college and to explore the potential of implementing an AI-driven decision support tool. Data Gathering Procedure The proponents relied on online surveys with the selected respondents. The survey was sent through email and can be accessed by using their LPU account. The questions for the survey were made based on the research objectives. The survey has categories depending on the respondents if either they are students, thesis professors/advisers and program chair. The respondents' privacy and confidentiality were strictly maintained throughout the data gathering process. Multiple Constraints Analysis The ATLAS project is influenced by several constraints that shape its development and deployment. These constraints are primarily technical, user-related, and regulatory. While other areas, such as environmental and health-related constraints, have no relevance to this system. First technical constraints include the system's dependency on hardware such as internet access, mobile devices and computers, which are critical for accessing and operating the system. Limited access to these resources, especially in areas with poor connectivity or insufficient hardware availability, could impact the system's functionality and user experience. A stable internet connection is	
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	essential, as the system cannot function or be accessed without it, potentially delaying research processes. Regarding data privacy constraints, the system handles sensitive information such as student, adviser, and administrative accounts, necessitating strict measures to protect this data. All personal information will be anonymized to ensure confidentiality, and access will be restricted to authorized personnel only, reducing the risk of unauthorized access or data breaches. These measures align with privacy laws and reinforce user trust in the system’s integrity and security. As for the constraints that are not applicable to this project, environmental constraints are not relevant since ATLAS is a software-based project, it has no direct interaction with physical environments or ecosystems. It uses existing digital infrastructure, resulting in no significant ecological impact. Similarly, health and safety constraints are not relevant, since the project is entirely digital, it does not involve physical risks to users. There are no requirements for emergency preparedness measures or occupational health precautions. Lastly, sustainability constraints, while sustainability in terms of long-term system usability is a consideration, there are no concerns regarding physical resource management or ecological sustainability, as the system does not rely on consumables System Development Process The proponents utilized the Scrum model for the development of ATLAS: Advanced Thesis Logistics and AI System for LPU-C. The Scrum model has 5	
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components that the proponents utilized. This includes Product Vision, Product Backlog, Planning Meeting, Sprint Backlog, and Deployment.

Figure 2. Scrum Model

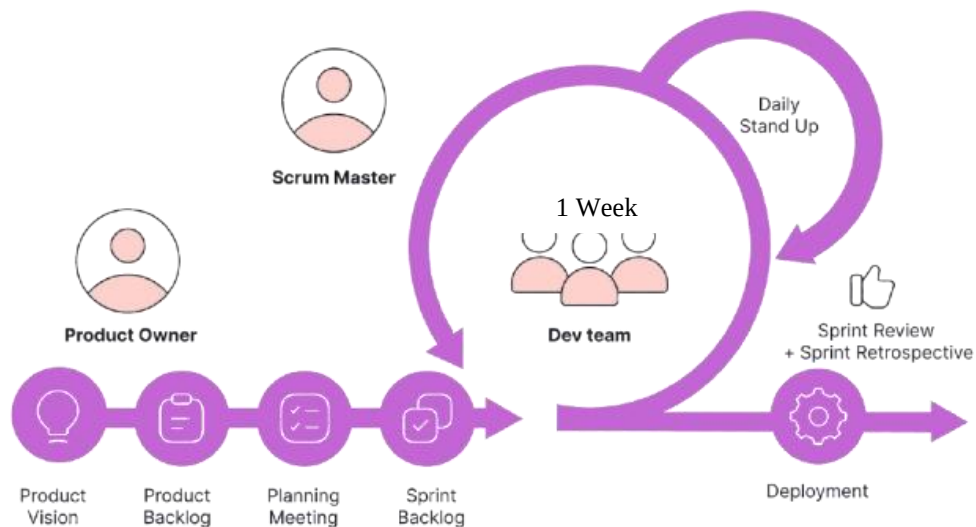


Figure 2 illustrates the process of Scrum Model. Below discusses each step pertaining to the actions of the life cycle:

Scrum Framework Overview

The Scrum model operates in short sprints typically lasting between 1-4 weeks. But for this study we intend to operate weekly. Each sprint consists of planning and development. Key roles in Scrum includes Product Owner, Scrum Master, and Development Team, each roles plays a crucial role ensuring the success of the project.

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	Roles: Product Owner: Defines the product vision, prioritizes the product backlog, and ensures value delivery. Scrum Master: Facilitates Scrum processes, removes blockers, and drives team improvement. Development Team: Builds, tests, and documents the product increment collaboratively. Artifacts and Events: Product Vision: The ultimate goal and purpose of the product. Product Backlog: A prioritized list of all tasks, features, and requirements for the product. Sprint Planning Meeting: A session to define the sprint goal and select backlog items for the sprint. Sprint Backlog: A focused subset of the product backlog for the sprint. Daily Standup: A quick meeting to align team progress and identify blockers. Sprint Review: A presentation of completed work for stakeholder feedback. Sprint Retrospective: A reflection meeting to identify and plan process improvements. Deployment: The release of the product increment to stakeholders or production.	
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Figure 3. Genetic Algorithm

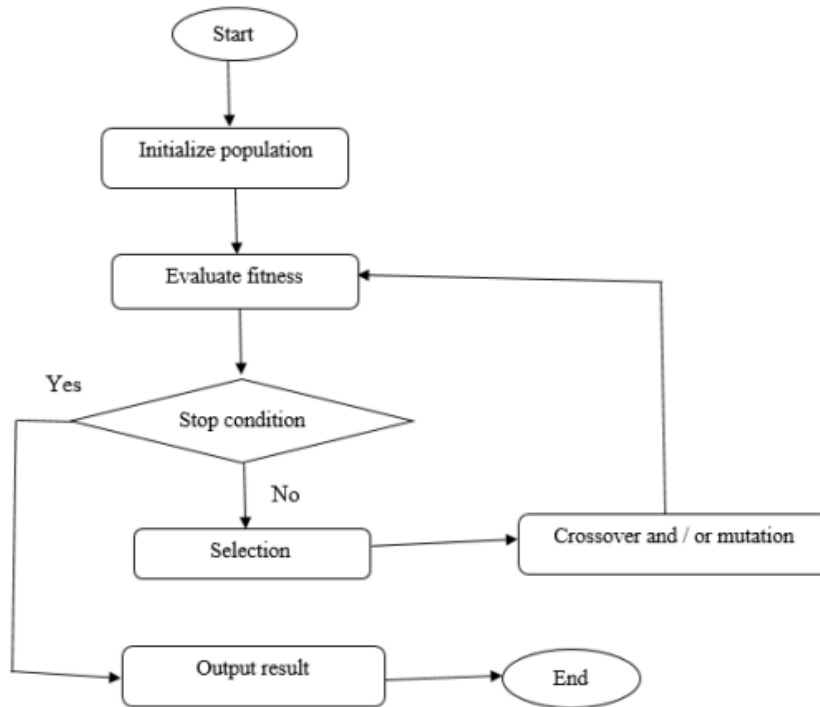


Figure 3 illustrates the flow of Genetic Algorithm. The algorithm starts by initializing the population. A population of potential solutions is randomly generated. Each chromosome encodes a candidate solution for the optimization problem, then the size of the population is predetermined. Next step is to evaluate the fitness. Each chromosome in the population is evaluated using a fitness function to determine how well it solves the problem. The fitness scores indicate the quality of a solution. Next step is the stop condition, the algorithm checks whether the termination condition is met. Common stop conditions include a specific number of generations has been reached, a satisfactory fitness level is achieved, and no significant improvement occurs over several iterations. If the

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	condition is met, the algorithm moves to output the result. If the condition is not met, then the next step is Selection. Selection chooses individuals (chromosomes) from the population based on their fitness scores. Higher fitness individuals are more likely to be selected for reproduction. Techniques like roulette wheel selection, tournament selection or rank-based selection are often used. Next step is the Crossover and Mutation. Crossover combines two parent chromosomes to produce offspring (new solutions). This mimics reproduction in nature and introduces new genetic material. Mutation randomly alters one or more genes in a chromosome. This helps maintain diversity and prevents the algorithm from converging prematurely on local optima. Next step is to evaluate the fitness of a new population. After crossover and mutation, the fitness of the new population is evaluated. Next is to repeat the process, the algorithm cycles back to the stop condition to determine if further iterations are needed. If not, it continues to the process of selection, crossover, and mutation. Next step is the output result, when the stop condition is met, the algorithm outputs the best solution found (ex. The chromosome with the highest fitness). Finally, the algorithm terminates after the optimal or satisfactory solution is reached.	
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Algorithm Simulation

Genetic Algorithms were designed/developed to simulate the process of natural evolution. GA's optimize populations of sequences. An initial population set can be created from a profile or at random. Based on their fitness, the algorithm modifies the population's sequences.

Setup

Parameters:

- **Teams:** Team A and Team B
- **Panelists:** P1, P2, P3, and P4
- **Rooms:** Room 1, Room 2
- **Time Slots:** 9:00 AM, 10:00 AM
- **Days:** December 11, 2024

Initial Population:

- **Population Size:** 2
- **Generations:** 2 (binabasa ko pa and inaaral)
- **Mutation Rate:** 10% (0.1)

Initial Population

Table 1. Individual 1 (Schedule)

Team	Panelists	Room	Time Slot	Day
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Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P2, P3, P4	Room 1	9:00 AM	December 11, 2024

Conflicts:

1. Both teams are scheduled in the same room at the same time.
2. Panelist P2 and P3 have overlapping schedules.

Fitness: -20 (e.g., -10 for each conflict).

Table 2. Individual 2 (Schedule)

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P4	Room 2	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

Conflicts:

1. Panelist P4 is double-booked between Team A and Team B.

Fitness: -10.

Selection

1. **Sort by Fitness:** Individual 2 is better (-10 vs. -20).
2. **Selected Individuals:** Top 50% are chosen. Here, both individuals are selected as the population size is small.

Crossover

- **Parent 1:** Individual 1
- **Parent 2:** Individual 2

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Crossover Point: After the first team's schedule.

Table 3. Child (New Schedule):

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 1	10:00 AM	December 11, 2024

Conflicts:

1. Panelist P3 has overlapping schedules.
2. Room 1 have been booked twice.

Mutation

Mutation Chance: Each team's schedule may change with a 10% chance.

- **Team A:** No mutation.
- **Team B:** Room changes to Room 2.

Table 4. Mutated Child

Team	Panelists	Room	Time Slot	Day
Team A	P1, P2, P3	Room 1	9:00 AM	December 11, 2024
Team B	P3, P4, P1	Room 2	10:00 AM	December 11, 2024

Conflicts:

- No conflicts after mutation.

Fitness: 0 (Perfect schedule).

Next Generation

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Population After 1st Generation:

1. Child: Fitness = 0
2. Best Individual from Previous: Individual 2 (Fitness = -10)

Final Generation

New Selection:

- Child with Fitness = 0 dominates.
- Another crossover or mutation may occur, but results will favor conflict-free schedules.

It will iterate depending on the set number of generation, the tuning of 100 population, 200 generations, and 0.1 mutation rate returned significant changes with least errors.

The input values will be sent as POST data to the run_scheduler.php file, which will also retrieve data from the database, including information about teams, users (panelists), and user schedules.

The algorithm will generate a randomized initial population of 100 solutions and iterate through 200 generations, using a learning rate of 0.1. Each team will be limited to a single schedule.

For each generation:

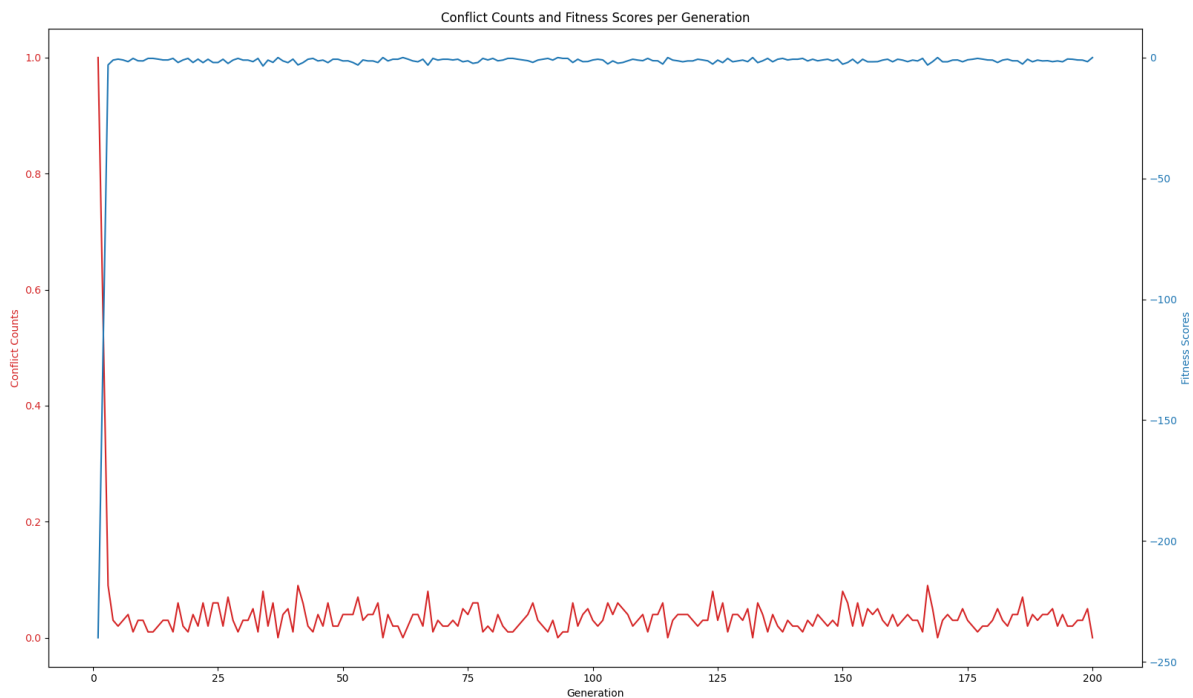
1. The algorithm computes the fitness score and counts conflicts for each individual in the population.

2. It calculates the average fitness and conflict count for the generation.
3. The solution with the least conflicts is selected, and genetic crossover is performed on its attributes (e.g., time slots, days, and rooms). These attributes are pooled and passed down to the next population.

The genes are then randomly assigned to the new population. If a newly generated gene assignment introduces more conflicts, the program reverts the changes and moves on to the next population member.

Finally, the population is sorted from the best to worst fitness scores, with the top individual representing the current best solution. Below is an example run.

Figure 4. Conflict Counts and Fitness Scores per Generation



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The graph shows the performance of a genetic algorithm (GA) over generations, with two metrics plotted:

1. **Conflict Counts (left y-axis, red line):** This likely represents the number of violations or conflicts (e.g., constraints in a scheduling problem) in the solution at each generation.
2. **Fitness Scores (right y-axis, blue line):** This represents how optimal or "fit" the solutions are (lower fitness scores could indicate better performance if minimizing is the goal).

Here's why the graph looks like this:

Initial Generations

- **Sharp drop in conflict counts:** In the early generations, the algorithm explores the search space and rapidly improves the population by eliminating many conflicts. This is typical as random solutions with high conflicts are refined through crossover, mutation, and selection.
- **Corresponding fitness improvement:** Fitness scores also improve quickly during these initial iterations, reflecting better solutions with fewer conflicts.

Stabilization Phase

- **Conflict counts oscillation:** After the initial rapid improvement, the red line stabilizes around a low value with slight fluctuations. This suggests the

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algorithm has reduced most conflicts but is struggling with local optimization and minor adjustments.

- **Fitness scores plateau:** Similarly, the blue line plateaus, indicating the population is converging to solutions that are near-optimal.

Reason for Oscillations

- **Genetic diversity:** Fluctuations in both metrics may result from mutations or crossovers introducing diversity into the population. This can momentarily worsen solutions before re-converging to better solutions.
- **Local optima:** The GA might be oscillating around a local optimum if global exploration is limited.

Observations

- The fitness scores and conflict counts don't seem to improve significantly after a certain point, which may indicate:
 - The algorithm has converged to a near-optimal solution.
 - Further refinement may require a hybrid approach, such as local search methods post-GA.

Pseudocode of Genetic Algorithm

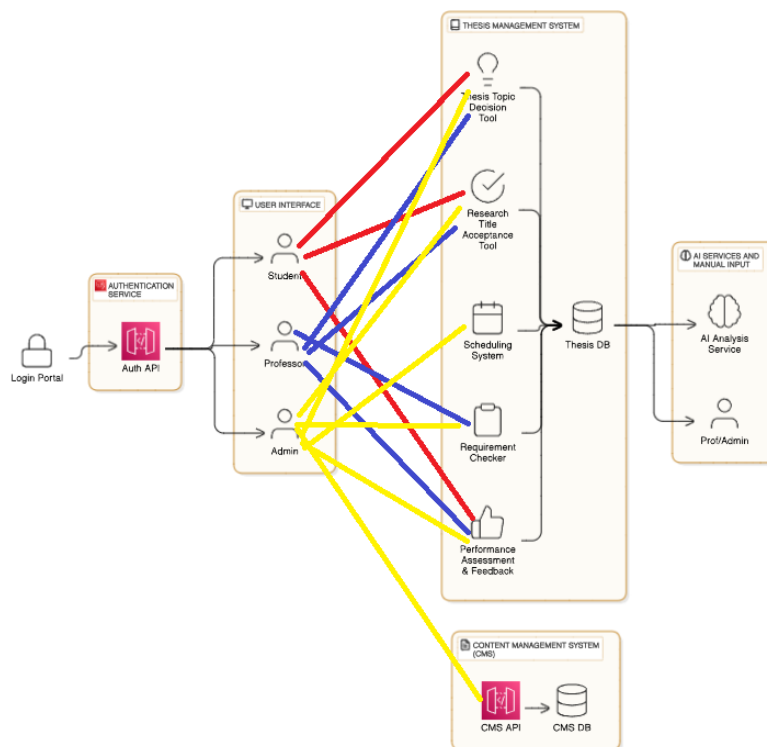
1. Start
2. # Step 1: Initialize
3. Enable error reporting
4. # Step 2: Handle POST Request

	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	<pre>5. If HTTP request method is POST: 6. # Retrieve data from the database 7. teams = fetchTeamsFromDatabase() 8. panelists = fetchPanelistsFromDatabase() 9. userSchedules = fetchUserSchedulesFromDatabase() 10. # Get user inputs 11. duration = getPostValue("duration") 12. rooms = getPostValue("rooms") 13. timeSlots = getPostValue("time_slots") 14. days = getPostValue("days") 15. # Validate inputs 16. If not isValidNumber(duration): 17. Throw Exception("Invalid duration") 18. # Step 3: Genetic Algorithm Initialization 19. population = [] 20. For i = 0 to populationSize: 21. schedule = createRandomSchedule(teams, panelists, rooms, timeSlots, days) 22. population.append(schedule) 23. # Step 4: Evolution Process 24. For generation in 1 to maxGenerations: 25. # Calculate fitness for each schedule 26. For each schedule in population: 27. schedule.fitness = calculateFitness(schedule) 28. # Track statistics</pre>	
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	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	<pre>29. trackStatistics(population) 30. # Sort population by fitness 31. population.sort(by="fitness", descending=True) 32. # Select top individuals for reproduction 33. selectedSchedules = selectTopHalf(population) 34. # Perform crossover and mutation to create new population 35. newPopulation = [] 36. While size(newPopulation) < populationSize: 37. parent1, parent2 = selectParents(selectedSchedules) 38. child = performCrossover(parent1, parent2) 39. mutate(child, mutationRate) 40. resolveConflicts(child) 41. newPopulation.append(child) 42. End While 43. population = newPopulation 44. End For 45. # Step 5: Save and Respond 46. bestSchedule = population[0] # Best schedule after all generations 47. If saveToDatabase(bestSchedule): 48. result = prepareSuccessResult(bestSchedule) 49. saveToJsonFile(result) 50. sendJsonResponse(result) 51. Else: 52. Throw Exception("Failed to save schedule") 53. Else:</pre>	
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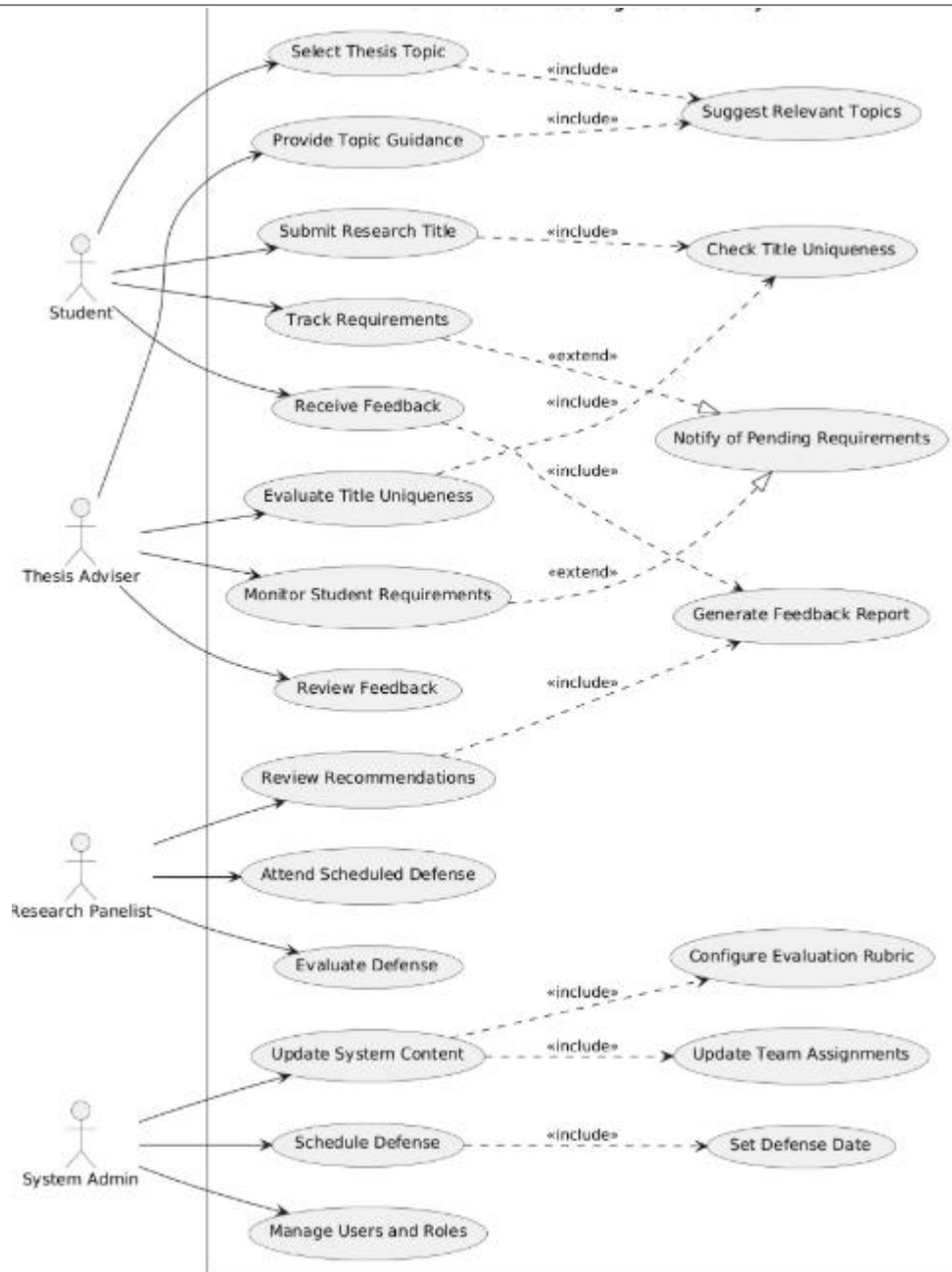
	LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE	
	<pre>54. Throw Exception("Invalid request method") 55. # Step 6: Error Handling 56. Catch Exception as e: 57. logError(e) 58. End</pre> The initialization phase of a Genetic Algorithm (GA) involves creating the initial population, where each individual represents a potential solution, such as a defense schedule for a team. This step is crucial as it provides the starting point for the algorithm's evolution. Fitness calculation follows, assessing how well each schedule meets predefined criteria, such as minimizing conflicts or adhering to user inputs. Schedules with higher fitness scores are considered better solutions, and this evaluation guides the GA toward progressively improving results. Selection and crossover are the next key steps in the GA process. The population is sorted by fitness in descending order, and the fittest individuals are prioritized for reproduction. This mimics natural selection, where only the strongest survive and reproduce. Crossover combines segments of two parent schedules to create new offspring, introducing genetic diversity while ensuring that the generated schedules remain valid. This process helps explore the solution space more effectively, leveraging the strengths of both parents. Mutation introduces variability by randomly altering schedule attributes, such as panelists, rooms, or times, with a small probability. This step helps the algorithm	
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Figure 7. Systems Architecture



Object Modelling

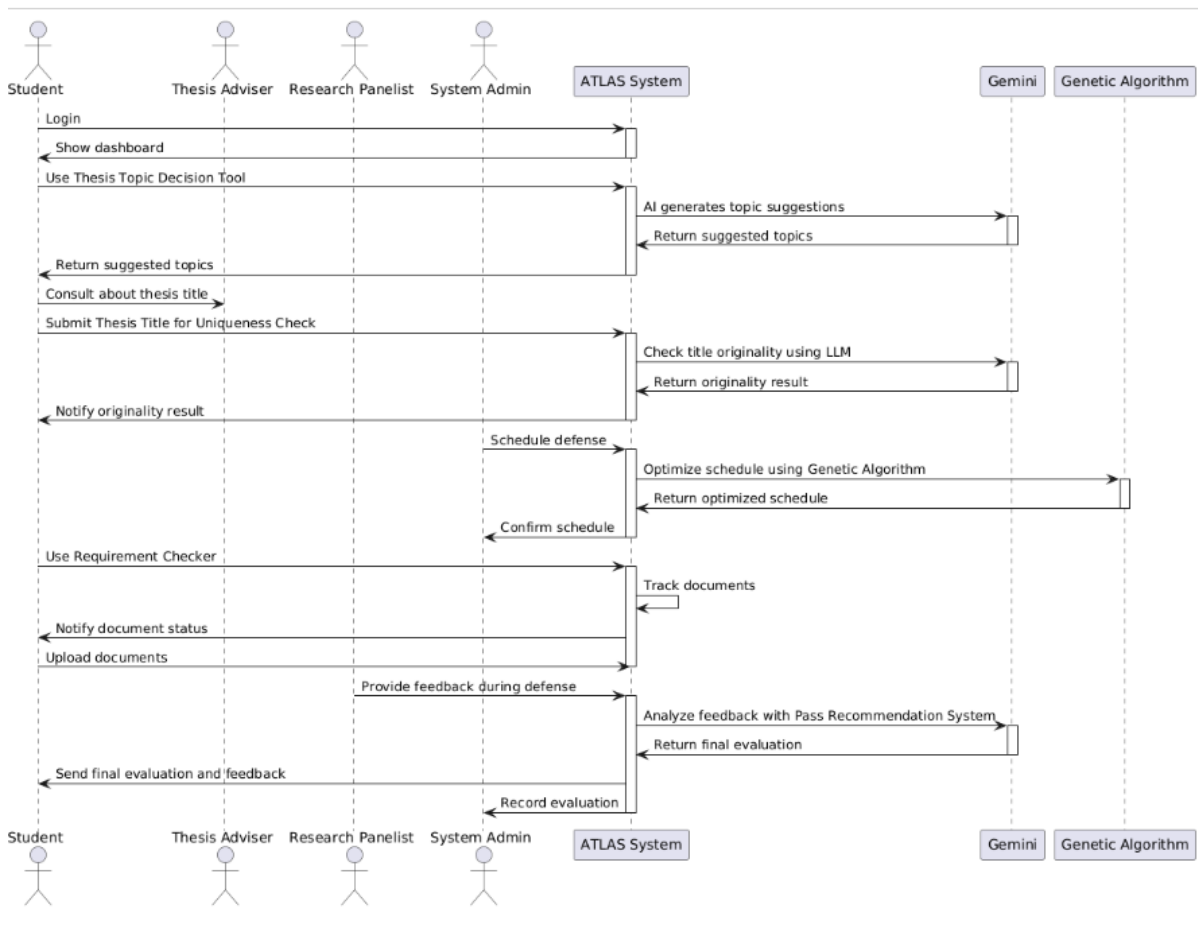
This portion of the design of the application demonstrates the expected client of the application and the capacities that were accessible for the clients to utilize.



Students initiate the process by selecting a topic with assistance from the Suggest Relevant Topics feature, ensuring relevance. They then submit a research title, which is verified for originality using Check Title Uniqueness. Students also track their document submissions through Track Requirements, with optional

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	reminders for incomplete tasks, and receive structured feedback after their defense through the Generate Feedback Report. Thesis Advisers guide students by refining suggested topics, reviewing title originality, and monitoring submission progress. They are alerted to any missing requirements and ensure the feedback provided to students is accurate and fair. Research Panelists focus on attending defenses, evaluating presentations, and finalizing assessments using system-generated feedback reports. System Admins oversee the defense scheduling process through Schedule Defense, which includes setting specific dates via Set Defense Date. They also manage user roles, update rubrics, and handle team or panel assignments. Additionally, Admins ensure system efficiency by addressing technical issues. Sequence Diagram This sequence diagram illustrates the interactions between actors (Student, Thesis Adviser, Research Panelist, System Admin) and systems (ATLAS System, Genetic Algorithm) for thesis management processes, including topic suggestion, title uniqueness checking, requirement tracking, defense scheduling, and evaluation, with specific tasks delegated to the Genetic Algorithm for optimized scheduling.	
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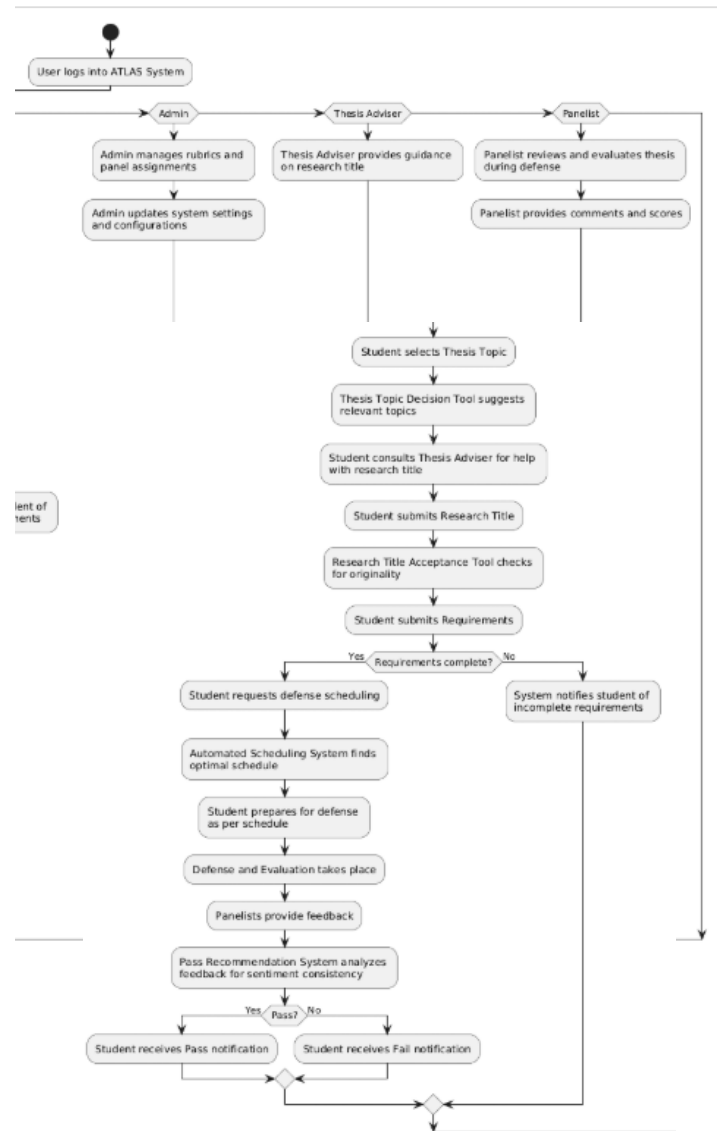
Figure 9. Sequence Diagram

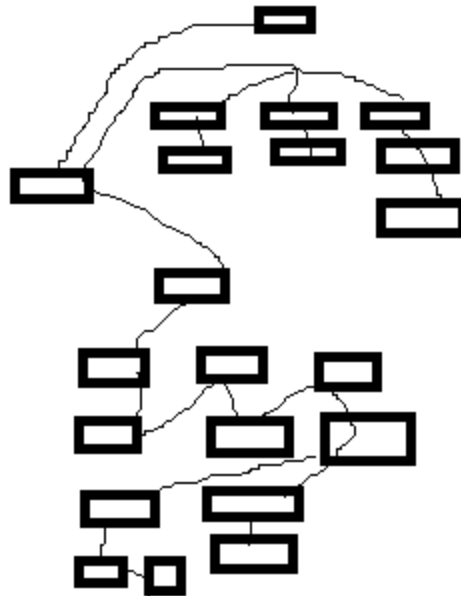


The process starts with the student logging in to use the Thesis Topic Decision Tool, where AI generates tailored topic suggestions. After consulting their Thesis Adviser, the student submits the chosen title to the system for an AI-powered originality check. The system evaluates the title and notifies the student of the results. The System Admin schedules the defense using optimization algorithms, while the student uploads required documents, tracked by a Requirement Checker. During the defense, the Research Panelist provides feedback, which the system

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	analyzes to generate final evaluations. The system then notifies the student and records results, ensuring efficiency and accuracy throughout the thesis process. Activity Diagram This activity diagram outlines the ATLAS system, where students consult their thesis advisers for research title guidance, submit requirements, and proceed to defense scheduling once all requirements are complete. Admins manage rubrics and configurations, advisers verify requirements and provide feedback, while panelists evaluate thesis defenses.	
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Figure 10. Activity Diagram





ILANO PAGAWA ITO NA GANTO FORMAT KUNG KAYA

Students begin by selecting a thesis topic and using the Thesis Topic Decision Tool for suggestions. They consult their thesis adviser for guidance on their research title before submitting it for originality verification via the Research Title Acceptance Tool. Afterward, students complete and submit all requirements. Once all requirements are fulfilled, students request a defense schedule, and the system's scheduling tool assigns an optimal time slot. During the defense, panelists review and evaluate the thesis, providing feedback and scores. The Pass Recommendation System analyzes this feedback to determine the final result, notifying the student of a pass or fail outcome. Admins manage rubrics, panel assignments, and system configurations, while thesis advisers guide students on

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research and provide feedback during the defense. Panelists focus on thesis evaluation and scoring. This process ensures efficient collaboration and streamlines thesis management.

Data Analysis

20 identify key challenges faced in the research process. A notable 65% of them have difficulty getting their research titles approved, and 70% struggle with selecting a suitable research topic. This emphasizes the necessity for AI tools such as the Research Title Acceptance Tool and Thesis Topic Decision Tool to improve these areas. What's more, 70% of students encounter issues with timely submissions of requirements, while 65% face confusion or delays regarding defense schedules. AI-driven solutions like a requirement tracker and an automated scheduling system could effectively resolve these challenges. When it comes to choosing research topics, 70% favor AI-powered suggestion tools, and 80% use online resources such as Google Scholar. To ensure the uniqueness of titles, 55% prefer using AI tools, although 50% also appreciate feedback from peers or mentors, indicating that a hybrid approach may be advantageous. In managing document submissions, 45% opt for a manual checklist, while 40% each prefer an AI-driven requirement tracker and guidance from faculty. This suggests that integrating AI with traditional methods could offer the most flexibility. Regarding evaluation tools, 55% find an AI-powered pass recommendation system quite useful, and 75% view a centralized system for managing defense requirements as highly beneficial. Opinions on AI-generated performance evaluations are varied; 45% find them very useful, while 25% still prefer traditional methods, indicating that AI should complement human evaluation rather than replace it.

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Ethical Considerations

In the development of ATLAS: Advanced Thesis Logistics and AI-System for LPU-Cavite COECSA Department, the following considerations are needed to ensure the protection of all stakeholders and the responsible use of AI technology.

1. Data Privacy and Confidentiality

- The proponents ensures that all the data/information gathered throughout the duration of this project will be handled with strict confidentiality. The proponents ensures that unauthorized users will not have access to this data. All participants will stay anonymous.

2. Informed Consent

- Before participating in this study, all participants are informed about the objectives and procedures involved. The proponents ensure that the participants fully understand, their rights and the voluntary nature of their involvement. Consent will be obtained through signed letters by the Dean of the department.

3. Intellectual Property and Plagiarism

- The system will respect and protect the intellectual property of both participants and the institution. Research titles, theses, and other academic content submitted by the institution and students will be safeguarded within

the system. The plagiarism detector will ensure originality and have no similar existing titles.

5. Cybersecurity and Protection from Harm

- ATLAS will be developed with strong cybersecurity protocols to prevent unauthorized access, data breaches, or system manipulation. Protection from harm also includes ensuring that the system does not expose participants to any form academic, emotional, or reputational harm through errors or misuse.

6. Bias and Fairness

- To address potential algorithmic bias in ATLAS, specific attention was given to ensure fairness in the Genetic Algorithm used for scheduling and Gemini LLM used in Thesis Topic Decision Tool, Research Title Acceptance Tool, and Pass Recommendation Tool. Actions were made to prevent any biases that might favor groups or lead to unfair outcomes.
- Transparency is also a guiding principle in this study. A clear documentation was provided on how the proponents gathered data, used the data in this study, and analyzed the data within ATLAS.

7. Accountability and Integrity

- The researchers and developers of ATLAS will maintain accountability throughout the system's design, deployment and operation. The research

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	team also ensures that ATLAS operates with the highest level of academic integrity, supporting fair evaluation practices. 8. Collaboration Ethics • ATLAS encourages collaboration between students, faculty, and the administrative staff while upholding ethical standards in all interactions. Contributions made by all participants during the systems development will be acknowledged. The researchers and developers of ATLAS ensures that the system will be used as a tool for academic growth while promoting responsible conduct in research and academic evaluation. These ethical considerations ensure that an IT thesis or capstone project is conducted responsibly, safeguarding the interests of individuals, society, and the broader tech community.	
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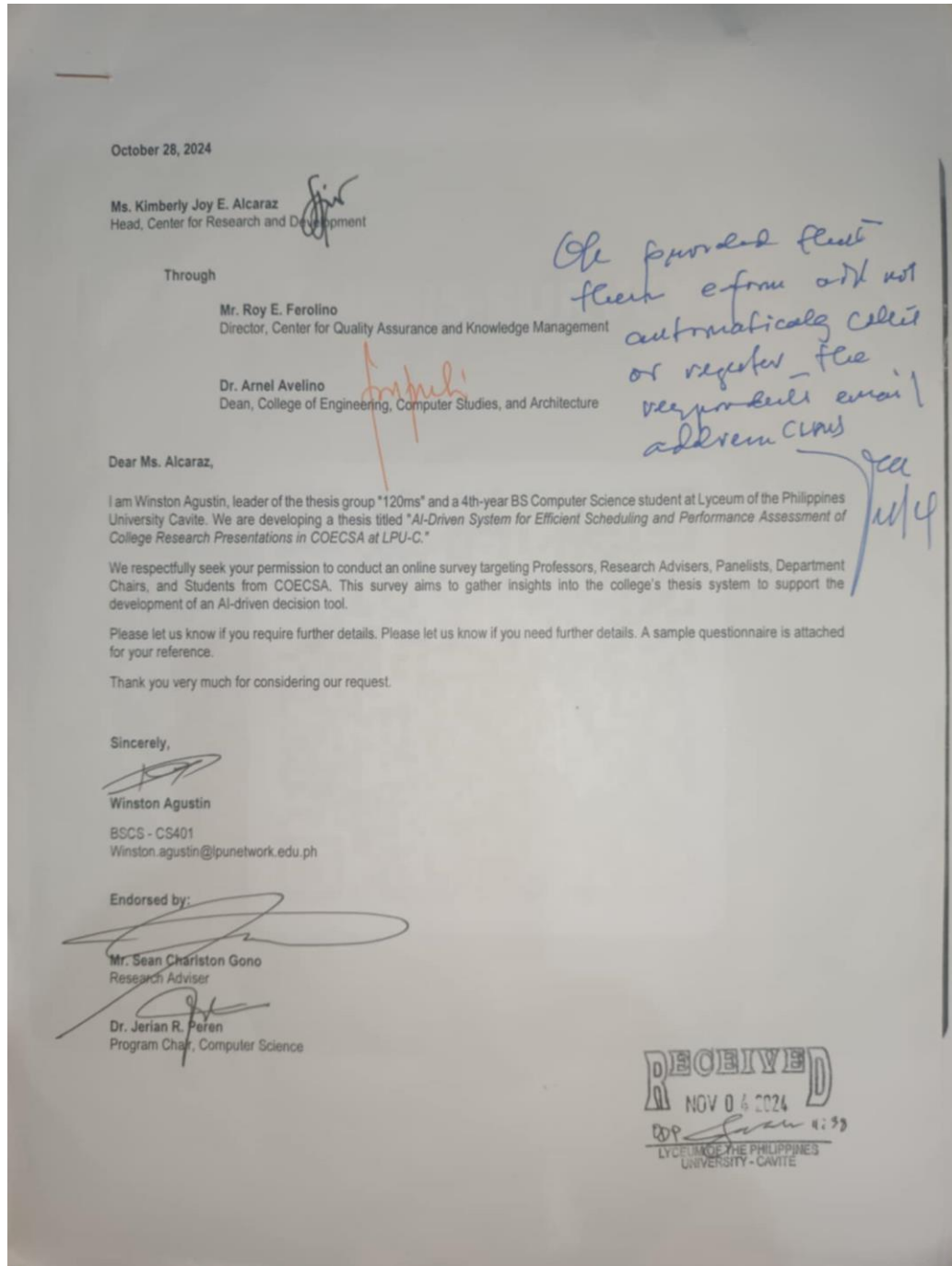
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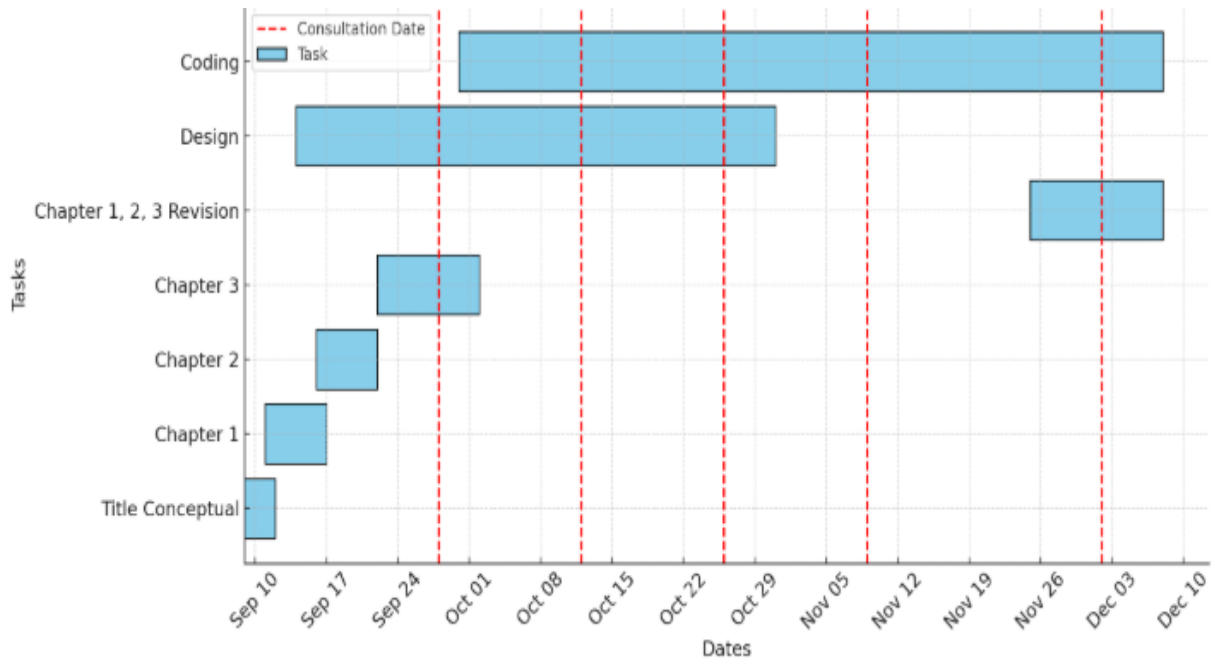
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APPENDICES

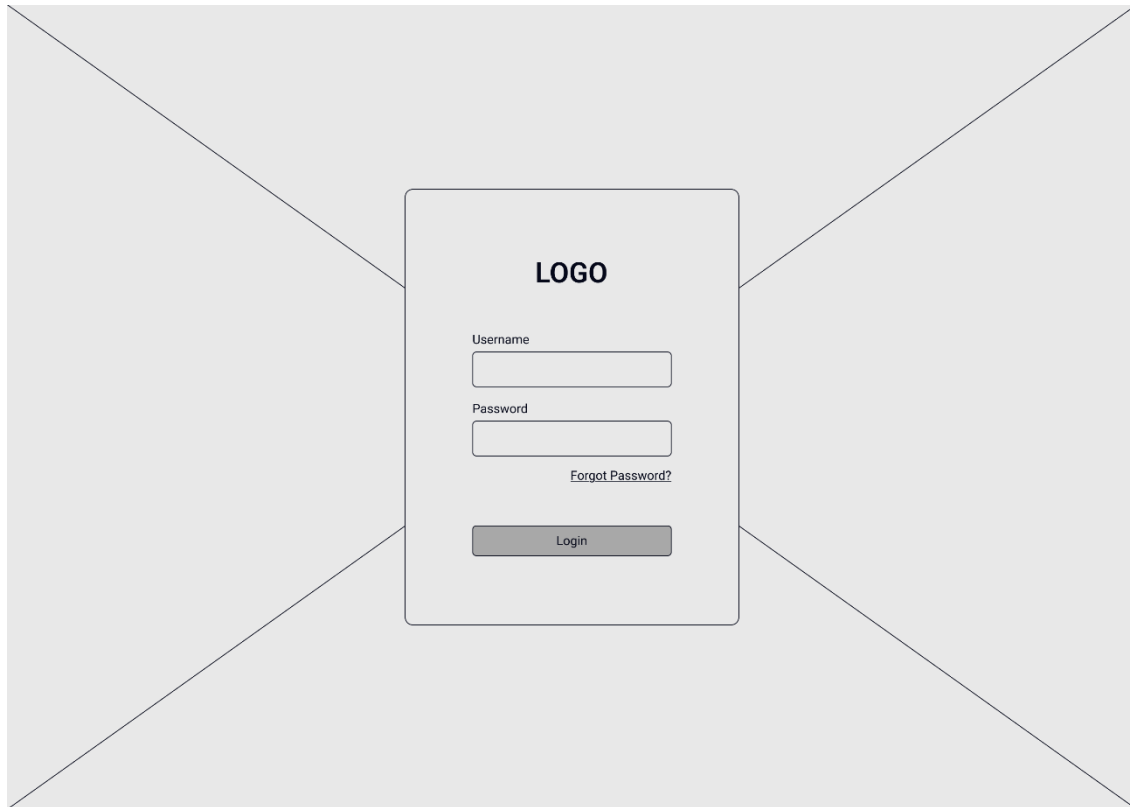
APPENDIX A LETTER OF REQUEST



**APPENDIX B
GANTT CHART**

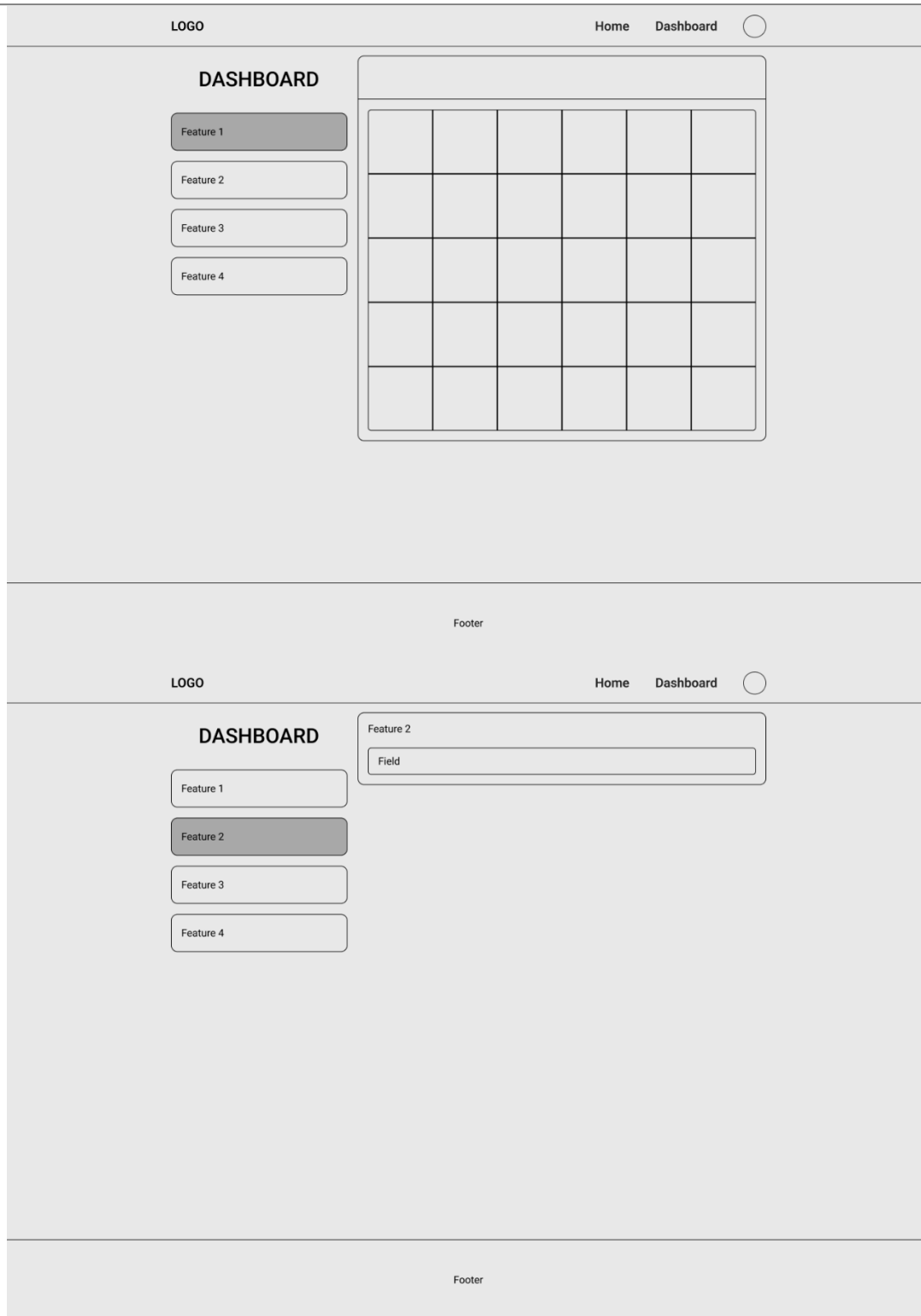


**APPENDIX C
WIREFRAME**



A wireframe of a login page. The page has a light gray background with a large, light gray rectangular area in the center. Inside this area, there is a white rectangular box with a thin gray border. At the top of this box is the word "LOGO" in bold. Below it are two input fields: the first is labeled "Username" and the second is labeled "Password". Below the password field is a link that says "Forgot Password?". At the bottom of the box is a button labeled "Login".

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LOGO Home Dashboard

DASHBOARD

Feature 1
Feature 2
Feature 3
Feature 4

Feature 3

Field
Field
Button

Footer

LOGO Home Dashboard

DASHBOARD

Feature 1
Feature 2
Feature 3
Feature 4

Feature 4

Requirement
Details
Details
Details
Button

Requirement
Details
Details
Details
Button

Requirement
Details
Details
Details
Button

Requirement
Details
Details
Details
Button

Footer



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LOGO

Home Dashboard

DASHBOARD

Button

Field

Button

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1

2

3

Dashboard feature 1

Dashboard feature 2

Dashboard feature 3

Dashboard feature 4

Dashboard feature 5

Dashboard feature 6

Dashboard feature 7

Dashboard feature 8

Dashboard feature 9

Dashboard feature 10

Footer

LOGO

Home Dashboard

DASHBOARD

Button

TH	TH	TH	TH
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Dashboard feature 1

Dashboard feature 2

Dashboard feature 3

Dashboard feature 4

Dashboard feature 5

Dashboard feature 6

Dashboard feature 7

Dashboard feature 8

Dashboard feature 9

Dashboard feature 10

Footer

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LOGO

Home

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Requirements

Defense Sched

Footer

LOGO

Home

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 2

Field

Footer

LYCEUM OF THE PHILIPPINES UNIVERSITY - CAVITE

LOGO

Home

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 3

Field

Field

Button

Footer

LOGO

Home

DASHBOARD

Feature 1

Feature 2

Feature 3

Feature 4

Feature 4

Requirement
Details
Details
Details

Button

Requirement
Details
Details
Details

Button

Requirement
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Button

Requirement
Details
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Details

Button

Footer

